

Epic 2: Visualizing and Analysing the Data

Story 1: Import the Required Python Libraries

Step 1: Open the Development Environment

Open Jupyter Notebook, Google Colab, or Visual Studio Code and create a new Python file or notebook.

Step 2: Import the Pandas Library

Import the Pandas library to load, organize, manipulate, and analyze the dataset using DataFrames.

Step 3: Import the NumPy Library

Import NumPy to perform numerical calculations and mathematical operations efficiently.

Step 4: Import the Matplotlib Library

Import Matplotlib to create charts such as line graphs, bar charts, histograms, and scatter plots.

Step 5: Import the Seaborn Library

Import Seaborn to create advanced statistical visualizations including heatmaps and box plots.

Step 6: Import Machine Learning Libraries

Import Scikit-learn modules including Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, Train-Test Split, One-Hot Encoder, Randomized SearchCV, Classification Report, Confusion Matrix, and F1 Score for model building and evaluation.

Step 7: Import Additional Libraries

Import any additional libraries required for handling imbalanced datasets and improving model performance.

Step 8: Execute the Import Statements

Run the import statements and verify that all libraries are loaded successfully without errors.

Step 9: Verify the Environment

Confirm that the Python environment is ready for data analysis and machine learning development.

Step 10: Prepare for Data Analysis

Proceed with dataset loading, preprocessing, visualization, feature engineering, model training, evaluation, and deployment.

Outcome

The Python environment is configured successfully, all required libraries are imported, and the project is ready for the complete Credit Card Approval Prediction workflow.